

Practical 3

About this unit

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Practice Lab Assignment

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Lab Assignment

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array)
- An integer representing the number of dimensions of the array
- A tuple representing the shape of the array
- An integer representing the total number of elements in the array

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

numpyarr...

```
1 import numpy as np
2
3 # Input rows and columns
4 r, c = map(int, input().split())
5
6 # Handle empty array case
7 if r == 0 and c == 0:
```

Average time: 0.004 s, Maximum time: 0.007 s, 2 out of 2 shown test case(s) passed, 10 out of 10 hidden test case(s) passed

Test case 1: Expected output: 3 4, 1 2 3 4, 5 6 7 8, 9 10 11 12, Actual output: 3 4, 1 2 3 4, 5 6 7 8, 9 10 11 12, [[1 2 3 4]]

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3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

- Addition (`matrix_a + matrix_b`)
- Subtraction (`matrix_a - matrix_b`)
- Element-wise Multiplication (`matrix_a * matrix_b`)
- Matrix Multiplication (`matrix_a * matrix_b`)
- Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

- The result of Addition.

matrixOp...

```
1 # Input matrices
2 print("Enter Matrix A:")
3 matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
4
5 print("Enter Matrix B:")
```

Average time: 0.007 s, Maximum time: 0.010 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: Expected output: Enter Matrix A: 1 2 3, 4 5 6, 7 8 9, Enter Matrix B: 1 2 3, 4 5 6, 7 8 9, Actual output: Enter Matrix A: 1 2 3, 4 5 6, 7 8 9, Enter Matrix B: 1 2 3, 4 5 6, 7 8 9

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3×3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
```

Average time: 0.007 s, Maximum time: 0.009 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: Expected output: Enter Array1: 1 2 3, 4 5 6, 7 8 9, Enter Array2: 4 5 6, 7 8 9, Actual output: Enter Array1: 1 2 3, 4 5 6, 7 8 9, Enter Array2: 4 5 6, 7 8 9

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate sequence
9 sequence = np.arange(start, stop, step)
10
11 # Print output in required format
12 print(sequence)
13
```

Average time: 0.004 s4.00 msMaximum time: 0.005 s5.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

- Arithmetic Operations
 - Compute the element-wise sum, difference, and product of the two arrays.
- Statistical Operations
 - Calculate the mean, median, and standard deviation of array A.
- Bitwise Operations
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A₁ OR B₁).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4     # Convert A and B to NumPy arrays
5     A = np.array(A)
6     B = np.array(B)
```

Average time: 0.005 s4.87 msMaximum time: 0.007 s7.00 ms1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1
Expected output: 1 2 3 4
Actual output: 1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Median of A: 2.5
Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12
Bitwise XOR: 4 6 7 12

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3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>
- After modifying the copy:

copyAnd...

```
1 import numpy as np
2
3 inputlist = list(map(int, input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
```

Average time: 0.003 s2.75 msMaximum time: 0.005 s5.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1
Expected output: 10 20 30 40 50 60 70 80
Actual output: 10 20 30 40 50 60 70 80

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2.5. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as space-separated integers as input from the user. Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching: Find the indices where search_value appears in array1 and print these indices.
- Counting: Count how many times count_value appears in array1 and print the count.
- Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.

arrayOpe...

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
```

Average time: 0.005 s5.50 msMaximum time: 0.007 s7.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1
Expected output: 1 1 1 2 2 2
Actual output: 1 1 1 2 2 2

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3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.

Operation...1import numpy as np23a = np.loadtxt("Sample.csv", delimiter=",", skiprows=1)4import numpy as np56a = np.loadtxt("Sample.csv", delimiter=",", skiprows=1)7Average time0.009 sMaximum time0.009 s1 out of 1 shown test case(s) passedTest case 1Expected outputAll student Details:[[301, 67, 77, 88]]Actual outputAll student Details:[[301, 67, 77, 88]]

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...1import pandas as pd23# Take inputs from the user to create a list of numbers4numbers = list(map(int, input().split()))56# Create a Pandas series from the list of numbers7Average time0.005 sMaximum time0.015 s3 out of 3 shown test case(s) passed3 out of 3 hidden test case(s) passedTest case 1Expected output1 2 3 4 5 6 7 8 9 10Mean of even and odd numbers:Odd: 5.0Even: 6.0dtype: float64Actual output1 2 3 4 5 6 7 8 9 10Mean of even and odd numbers:Odd: 5.0Even: 6.0dtype: float64